PDA navigation map

Sign in and start the shift

The person signs in, sees what there is to do and starts. The engine has already chosen the next task.

Smart Warehouse · PDA-01 · 100%
Welcome
Sign in to start working
Sign in
Use your username and password.
Username
E.g. 1042 or marta.silva
Password
Enter your password
Sign in

01 · Sign in
Identify who is using the PDA, with username and password.

PUT-AWAY · Marta Silva · online
Good morning, Marta
Work
Start tasks
Note GA-4471 · 3 pallets
Operations
Fix a location
Look up an item
Look up a location
Task · Home · Sign out

02 · Put-away home
Show the shift's work and get the person working with one tap; gives access to the look-ups and fixes they do on their own initiative.

Receive what arrives

The system does not know what is coming: when goods reach the dock, put-away says what arrived. That creates the receiving task, which the engine puts in the queue by priority. It is checked against the supplier note, goes into stock, and the put-away tasks go back to the engine.

Back · Arrival · online
What arrived?
The system does not know what is coming. Say who brought it and how many pallets.
Scan the note, if it has a barcode
Supplier
Castrol Portugal
Pallets
− 2 +
Log the arrival

04 · Log the arrival
Log the arrival: the system does not know what is coming, so this is where the goods enter the system.

Back · Arrival · online
Arrival logged at 10:05
Castrol Portugal · 2 pallets at the dock
Sent to the engine
Receiving task — 1 new
Where it is — Inbound dock
Check this arrival
Scan another arrival

05 · Arrival logged
Confirm the arrival is logged and the incoming task went to the engine.

PUT-AWAY · Marta Silva · online
Receive · Step 1 of 3
Scan the NOTE
The document that came with the goods, or the supplier's reference
At the inbound dock
2 pallets · Castrol Portugal
Aim at the note and press the trigger
Can't scan — type it

06 · Scan the supplier's delivery note
Start the receiving task. Identify the delivery by the supplier's note.

PUT-AWAY · Marta Silva · online
Receive · Step 2 of 3
Note GR-58213 · Castrol Portugal
Check what arrived
Expected
5W30 oil · 5 L — 40 pcs
W712 oil filter — 24 pcs
It's all here — confirm
The quantity doesn't match

07 · Check what arrived
Compare what was expected with what arrived.

PUT-AWAY · Marta Silva · online
Receive · Step 2 of 3
5W30 oil · 5 L · 40 expected
How many arrived?
− 36 +
4 pieces missing
Recorded as a receiving discrepancy. Receiving goes on.
Only what arrived goes into stock.
Record and continue

08 · The quantity doesn't match
Record how much actually arrived, when it isn't what was expected.

PUT-AWAY · Marta Silva · online
Receive · Step 3 of 3
Booked in at 10:18
GR-58213 · Castrol Portugal · 2 pallets
Into stock
5W30 oil · 5 L — 40 pcs
W712 oil filter — 24 pcs
Put-away tasks — 2 new
The 2 pallets stay at the dock until they are put away.
Start putting away
Scan one more arrival

09 · Received
Confirm it is in stock, and that the put-away tasks went to the engine, which sets their priority in the queue.

Put away a pallet

Take the pallet from the dock to the location the engine suggests, and confirm by scanning the item and the location.

PUT-AWAY · Marta Silva · online
GA-4471 · pallet 1 of 3 · Step 1 of 4
Scan the ITEM
The label on the box or pallet in front of you
Expected
5W30 oil · 5 L
40 pieces · from the inbound dock
Aim and press the trigger
Can't scan — type it

10 · Scan the item
Confirm that the pallet in front of the person is the one in the task.

PUT-AWAY · Marta Silva · online
GA-4471 · pallet 1 of 3 · Step 2 of 4
Go to · Suggestion 1 of 3
60300301
Zone 30 · Column 03 · Shelf 01
Why here
It moves fast — 34 a day — and is 8 m from the dock. Too heavy to go higher.
Route: leave the dock for zone 30, column 03 on the right.
Scan the location
It's taken
Another problem

11 · Go to suggestion 1
Say where to take the pallet, and why.

PUT-AWAY · Marta Silva · online
GA-4471 · pallet 1 of 3 · Step 2 of 4
60300301 was recorded as taken
The manager will look into why the system thought it was free.
Go to · Suggestion 2 of 3
60300401
Why here
It is the next closest to the dock that takes the weight.
Scan the location

12 · Number 1 was taken
Record the location as taken and move on to the next suggestion.

PUT-AWAY · Marta Silva · online
GA-4471 · pallet 1 of 3 · Step 3 of 4
Scan the LOCATION
The label on the shelf where you placed the pallet
Should be
60300401
When you scan, the system checks again that the location still fits
Can't scan — type it

13 · Scan the location
Confirm, by scanning, that the pallet is in the right location.

PUT-AWAY · Marta Silva · online
GA-4471 · pallet 2 of 3 · Step 4 of 4
Pallet put away in 60300401
40 × 5W30 oil · recorded under your name at 10:14
Stock status
Before — Waiting for put-away · dock
Now — Stored · 60300401
This note
Pallets put away — 2 of 3
Left — 1 pallet
When the last one is put away, the note closes and the engine gives you the next.
Next pallet
See the task

14 · Put away
Confirm the record and move on to the next task.

When put-away goes wrong

The detours — location taken, doesn't fit, damaged item, label that won't scan. Everything is recorded and the manager sees it.

PUT-AWAY · Marta Silva · online
Put away pallet · Step 2 of 4
60300301
What's wrong?
Location taken
It doesn't fit
Can't reach the location
Damaged item
Unreadable label
Other reason — type it

15 · What's wrong?
Say, with one tap, what is stopping the put-away.

PUT-AWAY · Marta Silva · online
Put away pallet · Step 1 of 4
Type the ITEM
The label won't scan. Type the reference on the box.
Item reference
OL5W30-5
Recorded as typed in
Confirm

16 · Type the item
Identify the item when the label won't scan.

PUT-AWAY · Marta Silva · online
Put away pallet · Step 3 of 4
Type the LOCATION
The eight digits on the shelf label
Location code
60300401
Recorded as typed in. If the label keeps failing, the system will advise renewing it.
Confirm

17 · Type the location
Identify the location when the label won't scan; counts towards advising a new label.

Report a problem

Any role reports what is stopping them: it is recorded with who said it and when, and the manager is told.

PUT-AWAY · Marta Silva · online
Put away pallet · Step 2 of 4
Damaged item
Recorded at 10:16 under your name
The manager has been told. The pallet stays where it is until they decide.
Next task

18 · Problem recorded
Confirm the problem is recorded and the manager has been told.

Fix a location (task)

The engine found stock in the wrong place and sends someone. It comes out of the source and goes into the destination, in a single record.

PUT-AWAY · Marta Silva · online
Fix a location · Step 1 of 2
Scan the SOURCE
Should be
60300302
Take
6 × 5W30 oil · 5 L
They are in the wrong place
Scan the location, then the item
It's no longer there

19 · Take from the source
Confirm the location and the item that come out.

PUT-AWAY · Marta Silva · online
Fix a location · Step 2 of 2
Goes to
60300401
Zone 30 · Column 04 · Shelf 01
Why here
It already holds the same item — it joins what is there.
Scan the location where you put it

20 · Put in the destination
Say where the stock goes, and confirm by scanning.

PUT-AWAY · Marta Silva · online
Fix a location · Step 2 of 2
6 × 5W30 oil moved
From 60300302 to 60300401 · at 10:22
Next task

21 · Fixed
Confirm the stock has moved, in a single record.

Fix what is in a location

On their own initiative: whoever sees a location doesn't match the system fixes it right there. Nobody counts.

Back · Fix a location · online
Scan the LOCATION
The label of the location where what's there doesn't match the system
Aim and press the trigger
Can't scan — type it

22 · Scan the wrong location
Identify the location that doesn't match the system.

Back · Fix a location · online
60300302
The system says
5W30 oil · 5 L — 12 pcs
How many are there?
− 6 +
Record the correction

23 · What's there?
Say how many pieces are actually in the location.

Back · Fix a location · online
60300302 fixed
5W30 oil from 12 to 6 · at 10:40
The manager sees the difference in the shift report.
Back to home

24 · Correction recorded
Confirm the correction; the manager sees the difference.

Take the empty pallet away

The pallet that became empty leaves the location and goes to the yard; the location is freed.

PUT-AWAY · Marta Silva · online
Empty pallet · Step 1 of 2
Take it to the YARD
The empty pallet from 60100201
Pallet yard
Next to the inbound dock
Scan the pallet yard label

25 · Take it to the pallet yard
Take the empty pallet to the yard, and confirm by scanning its label.

PUT-AWAY · Marta Silva · online
Empty pallet · Step 2 of 2
Empty pallet in the yard
60100201 became free at 10:45
Next task

26 · In the pallet yard
Confirm the location is now free.

Look up

Find out where an item is, or what is in a location, without changing anything.

Back · Look up an item · online
Scan the item, or type the reference
5W30 oil · 5 L
Where it is
60300401 — 40 pcs
60300301 — 12 pcs
Inbound dock — 36 to put away

27 · Look up an item
See where an item is, and how much is in each place.

Back · Look up a location · online
Scan the location
60300401
Zone 30 · Column 04 · Shelf 01
Item — 5W30 oil · 5 L
Quantity — 40 pcs
Last in — 10:14 · Marta Silva
Isn't this what's there? Fix it here.
Doesn't match — fix it

28 · Look up a location
See what is in a location and, if it doesn't match, fix it.

Starting the picking shift

Picking signs in, sees the round the engine gave them and starts. The batch fills itself and an order goes in whole.

PICKING · Rui Costa · online
Good morning, Rui
Work
Start picking
Now: order G-24187 · 9 lines
Operations
Look up an order
Look up an item
Look up a location
Task · Home · Sign out

29 · Picking home
Show the shift's round and get the person working with one tap; gives access to the look-ups.

Back · Picking · online
Your next task · one at a time
The engine gave you this order. When you close it, it gives you the next.
Now
Order G-24187 · Auto Peças Norte · 9 lines
Why: the DPD truck closes at 11:30
The stops are in zone 20, close to where you finished
In the warehouse
Orders waiting — 12
Being picked right now — 3
The engine shares them out by what the PDAs are doing. Nobody picks, and nobody takes someone else's order.
Start work

30 · Your next task
Mostrar a volta que o motor deu agora, e porque é que é esta.

Pick an order

Picking goes through the round of the warehouse with the batch of orders, and leaves them on the staging shelf.

PICKING · Rui Costa · online
Your order
G-24187 · Auto Peças Norte
Shelf — PP-DPD-03
Start picking

31 · Your order, stop by stop

PICKING · Rui Costa · online
Stop 4 of 9 · G-24187
Go to
60200201
Take
6 × W712 oil filter
For G-24187 · the stock is already reserved for you
Next — 60200502 · 5 m
Scan the LOCATION, then the ITEM
Short quantity
Type it
Another problem

32 · Stop: scan the location
Take the person to the location and say what to pick.

PICKING · Rui Costa · online
Stop 5 of 9 · G-24187
Only 2 of the 6 are here
Record what you took. The order stays incomplete and does not ship without the rest.
60200502
Ordered — 6
Taken — 2
Missing — 4
The 2 you took stay on the order. The difference is recorded on this location and the manager sees the signal — moving stock is their call.
Record and continue

33 · Short quantity
Registar o que se tirou e deixar a diferença assinalada na posição.

PICKING · Rui Costa · online
End of the order · 9 of 9
Take it to the staging shelf
PP-DPD-03
For the DPD truck · 11:30 · next to the outbound dock
G-24187 — complete · 9 parcels
Incomplete, it would wait on the same shelf: the engine hands it out again as soon as the stock comes in.
Scan the location of the shelf where you place it

34 · Take it to the staging shelf
Deixar a guia na prateleira de preparados, e dizer onde ficou.

PICKING · Rui Costa · online
End of the order · 9 of 9
Order closed at 11:02
G-24187 · complete · 9 parcels
You left
Where — PP-DPD-03
Truck — DPD · 11:30
Shipping takes it when the truck arrives. Incomplete, it would not reach shipping: it stays on the shelf, and the engine hands it out again when the stock comes in.
Next task

35 · Order closed
Confirmar que a guia ficou na prateleira e que a tarefa acabou.

When picking goes wrong

The detours of the round: nothing in the location, less than the order asks for, a different item, a label that will not scan — and leaving the shift with orders still open.

PICKING · Rui Costa · online
Stop 4 of 9 · G-24187
60200201
What's wrong?
Nothing in the location
Less than the order asks for
A different item is there
Unreadable label
Damaged item
Other reason — type it

36 · What's wrong?
Say, with one tap, what is stopping the pick.

PICKING · Rui Costa · online
Stop 4 of 9 · G-24187
Type the LOCATION
The eight digits on the shelf label
Location code
60200201
Recorded as typed in. If the label keeps failing, the system will advise renewing it.
Confirm

37 · Type the location
Identify the location when the label won't scan; counts towards advising a new label.

Back · Sign out · online
You have order G-24187 half done
5 lines picked, 4 still to pick
If you leave now
What you have taken — take it to PP-DPD-03
The 4 left — go back to the engine
The DPD truck — 11:30
Nothing goes back to the racks: what is on the trolley goes to this order's slot, and the lines are recorded. Whoever picks it up carries on at stop 6 and adds the rest in the same place. The manager sees who left an order open.
Carry on with the order
Leave anyway

38 · Leaving with the order half done
Dizer onde fica o que já se recolheu, antes de a pessoa sair com a guia a meio.

Look up (picking)

See how an order is doing, where an item is, or what is in a location, without changing anything.

Back · Look up an order · online
Scan the order or type its number
G-24187
Auto Peças Norte · DPD truck · 11:30
Status — complete

39 · Look up an order
See how an order is doing: what is missing, where it is and which truck it goes on.

Back · Look up an item · online
Scan the item, or type the reference
W712 oil filter
W712 · zone 20
Where it is
60200201 — 24 pcs
60200502 — 6 pcs
Reserved for orders — 12 pcs

40 · Look up an item
See where an item is, and how much is in each place.

Back · Look up a location · online
Scan the location
60200201
Zone 20 · Column 02 · Shelf 01
Item — W712 oil filter
Quantity — 24 pcs
Last out — 9:40 · Rui Costa
Isn't this what's there? Fix it here.
Doesn't match — fix it

41 · Look up a location
See what is in a location and, if it doesn't match, fix it.

Starting the shipping shift

Shipping signs in, sees the shift trucks by closing time and starts loading.

SHIPPING · Ana Lopes · online
Good morning, Ana
8 am shift · the engine gives you one order at a time
Work
Start loading
Now: order G-24187 · 9 parcels · DPD
Operations
Today's trucks
Look up an order
See the staging shelf
Task · Home · Sign out

42 · Shipping home
Show the shift trucks and get the person loading with one tap; gives access to the look-ups.

Back · Shipping · online
Your next task · one at a time
The engine gave you this order. When you load it, it gives you the next.
Now
Order G-24187 · 9 parcels · DPD truck
Why: the truck closes at 11:30 and is already at the dock
It is the next order for the same truck: you are already there
In the warehouse
Orders waiting — 7
Being loaded right now — 2
The engine shares them out by what the PDAs are doing. Nobody picks, and nobody takes someone else's order.
Start work

43 · Your next task
Mostrar o camião que o motor deu agora, e porque é que é este.

Look up (shipping)

See how an order is doing, and what is on the staging shelf.

SHIPPING · Ana Lopes · online
Today's trucks
By closing time
DPD — 11:30
1 order on the shelf · 0 at the dock · Tight
GLS — 2 pm
1 complete · 1 incomplete on the shelf
Transportes Vilar — 17:30
4 orders to pick
Back to home

44 · Today's trucks
Ver os camiões por hora de fecho e o risco de cada um; não é por aqui que se escolhe trabalho.

From the shelf to the truck

Shipping checks the parcels, takes them to the dock and loads the truck.

SHIPPING · Ana Lopes · online
G-24187 · DPD · Step 1 of 4
On the shelf · PP-DPD-03 ✓
Scan each PARCEL
● Box 1 · filters — V1/9
● Box 2 · oils — V2/9
● Box 3 · oils — V3/9
○ Box 4 · brake pads — V4/9
○ … 5 more parcels — V5–V9
3 of 9 scanned
A parcel is missing
Another problem

45 · Check parcel by parcel
Confirm each parcel before it leaves the shelf.

SHIPPING · Ana Lopes · online
G-24187 · DPD · Step 2 of 4
9 of 9 parcels checked
The order is complete.
Take it to the dock
CAIS-2
Where the DPD truck is backing in
Scan the label of the dock where you place it
Can't scan — type it

46 · Take it to the dock
Take the parcels to the truck's dock.

SHIPPING · Ana Lopes · online
G-24187 · DPD · Step 3 of 4
Scan while loading
On this truck
G-24187 · yours — 9/9 loaded
G-24181 — 4/4 loaded
G-24185 — 1/3
Scan each parcel as it goes onto the truck
When the last order goes up, the dispatch closes by itself. Closing before that is the call of whoever is at the dock, and the PDA asks first. Transport paperwork: on the shipping desk computer.
Order loaded
Close the dispatch

47 · Load and close the dispatch
Confirm each parcel that is loaded, and close the dispatch.

DPD dispatch · 11:30 · Step 4 of 4
DPD dispatch closed at 11:28
3 orders · 22 parcels loaded
Recorded
G-24187 — 9 parcels
G-24181 — 4 parcels
G-24185 — 9 parcels
The transport documents print on the shipping computer.
Next task

48 · Dispatch closed
Confirm the dispatch is closed and what was loaded.

When shipping goes wrong

A parcel missing from the shelf, the truck has not arrived, a label that will not scan — and leaving the shift with a dispatch still open.

SHIPPING · Ana Lopes · online
G-24187 · DPD · Step 1 of 4
PP-DPD-03
What's wrong?
A parcel is missing from the shelf
Damaged parcel
The truck has not arrived
Unreadable label
It doesn't fit in the truck
Other reason — type it

49 · What's wrong?
Say, with one tap, what is stopping the loading.

G-24187 · DPD · Step 1 of 4
Parcel V4/9 is missing
If it leaves like this
The order — stays incomplete
What goes — 8 of 9 parcels
The rest — waits for the next truck
Record and continue
See the order

50 · A parcel is missing
Record the missing parcel and say what happens to the order.

G-24187 · DPD · Step 2 of 4
Type the DOCK
The number of the dock the truck is backing into
Dock
CAIS-2
Recorded as typed in
Confirm

51 · Type the dock
Identify the dock when the label won't scan.

Back · Sign out · online
You have order G-24187 half done
5 parcels on the truck, 4 still on the shelf
If you leave now
The 5 already loaded — stay recorded
The 4 left — go back to the engine with the order
The DPD dispatch — stays open
No parcel comes back off the truck: the ones already loaded stay there and recorded. Whoever picks the order up carries on at parcel 6, and the truck waits at the dock until the dispatch closes. The manager sees who left an order open.
Carry on loading
Leave anyway

52 · Leaving with the order half done
Dizer o que acontece aos volumes já carregados, antes de a pessoa sair com a guia a meio.

Look up (shipping)

See how an order is doing and what is on the staging shelf.

Back · Look up an order · online
Scan the order or type its number
G-24187
Auto Peças Norte · DPD truck · 11:30
Where it is — PP-DPD-03
Goes out — today · 11:30 · DPD
Status — incomplete

53 · Look up an order
See how an order is doing: what is missing, where it is, and when it is due out.

Back · Shelf · online
Scan the shelf location
PP-DPD-03
On the shelf
G-24187 — 9 parcels
G-24181 — 4 parcels
G-24185 — 9 parcels

54 · See the staging shelf
See what is on the shelf and which truck each order goes on.

The manager

The manager sees how the shift is going, reads the signals, assigns tasks with a written reason and sees the whole warehouse.

MANAGER · Paulo Mendes · online
Good morning, Paulo
8 am shift · 47 tarefas feitas
Work
See the shift
Figures and signals
Operations
A truck has arrived
Insights
What is going on
Assign a task
Warehouse · Home · Sign out

55 · Manager home
Show the manager how the shift is going and give access to what only they do.

MANAGER · Paulo Mendes · online
8 am shift
So far · 10:20
47 tasks done
6%
3 locations taken
1 incomplete order
Signals · See all (5)
Map — 60300301 refused 3 times — The system thought it was free. Order a fix?
Stock — 60200502 has 2 open corrections — Stock may be leaving unrecorded
Arrival — Logged 14 min after it pulled in — The dock was tied up for that long, and the engine did not know.

56 · Shift report
Show the manager the shift figures and the signals that need attention.

Back · Assign · online
Task
Note GA-4471 · 3 pallets
To whom
Marta Silva
Why
Está na coluna ao lado
The engine records the pattern of assignments and proposes changes; but it is management who changes the rule.
Assign

57 · Assign a task
Give a task to a person, with the reason written; the engine learns from it.

Back · Now · online
Put-away
Marta Silva — putting away 60300401
Orders waiting — 4
At the dock — 2 pallets to receive
Picking
Rui Costa — G-24187 · stop 4 of 9
Orders still open — 1
Shipping
Ana Lopes — G-24187 · loading
Closes at — 11:30
Assign a task

58 · What is going on
See the whole warehouse: who is working, what is missing and what goes out today.

Back · Done · online
Task assigned to Marta Silva
With the reason written · at 10:22
She gets it as her next one, and the engine learns from the assignment. A logged arrival is born the same way: it becomes a task and goes to whoever finishes first.
Back to home

59 · Done
Confirm what the manager ordered, and say where it lands.

Back · All operations · online
Work as — recorded under their name
Put-away · Picking · Shipping
Operations
Arrival · Fix · Item
Position · Order · Shelf

60 · All operations
Give the manager everything: work as any role, and all of their operations.

Back · Insights · online
6%
12 min
Locations refused, by zone
Zone 30 — 7
Zone 20 — 3
Zone 10 — 1
What the engine learned
The manager's assignments — 4 this week
Pattern — oils late in the morning
The engine proposes changing the order of the rule. It is management who changes the rule.

61 · Insights
Show what the data says: where time is lost, what fails and what the engine learned.

When the connection drops

The PDA works connected to Robgest. With no connection it says it stopped, and shows what was saved.

PICKING · Rui Costa · NO CONNECTION
No connection
Since 10:22
The PDA is not connected to the server
You cannot start or close anything until the connection is back.
Saved
Stop 3 of 18 — confirmed at 10:21
Stop 4 of 9 — not confirmed
Move towards the main aisle. When the connection is back, you resume at stop 4.
Batch figures: from the last connection, at 10:21
Try again

62 · No connection
Say the PDA isn't connected, and what was saved.

From the shelf to the truck

Shipping checks the parcels, takes them to the dock and loads the truck.

SHIPPING · Ana Lopes · online
G-24187 · DPD · Step 3 of 4
Close the DPD dispatch?
2 parcels of G-24185 are missing. If you close it, they go on the next truck and the order stays incomplete.
Close it anyway
Back to loading

63 · Close the dispatch?
Perguntar antes de fechar a saída, que é o que não tem volta.

Fix what is in a location

On their own initiative: whoever sees a location doesn't match the system fixes it right there. Nobody counts.

PUT-AWAY · Marta Silva · online
Back · Fix · online
60300302
Fix 60300302?
The system says 12 and you are about to record 6. Stock drops by 6, and the manager sees the difference in the report.
Fix
Back to counting

64 · Fix the stock?
Perguntar antes de mudar o stock de uma posição, que ninguém desfaz.

Look up

Find out where an item is, or what is in a location, without changing anything.

Back · Look up an item · online
5W30 oil · 5 L
60300401 — 40 pcs
60300301 — 12 pcs
A look-up changes nothing. The work stays where it was.
You have a new task: note GA-4471

65 · Notice: new task
Dizer que o motor deu trabalho novo, sem interromper o que se está a fazer.

When put-away goes wrong

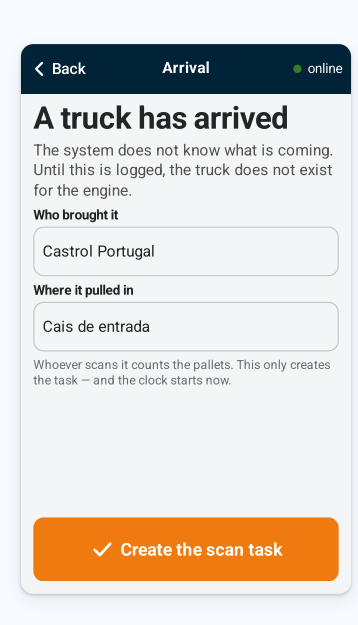
The detours — location taken, doesn't fit, damaged item, label that won't scan. Everything is recorded and the manager sees it.

Back · Sign out · online
You have note GA-4471 half done
2 of the 3 pallets are still to put away
If you leave now
The pallet already put away — stays done
The 2 left — go back to the engine
Where they are — goods-in dock
Each pallet is recorded the moment it is put away, so no work is lost. The engine gives the remaining ones to the next person, and the manager sees who left a note open.
Carry on with the order
Leave anyway

66 · Leaving with a note half done
Avisar que a guia fica a meio, e dizer o que acontece às paletes já arrumadas.

The manager

The manager sees how the shift is going, reads the signals, assigns tasks with a written reason and sees the whole warehouse.



A truck has arrived

The system does not know what is coming. Until this is logged, the truck does not exist for the engine.

Who brought it: Castrol Portugal

Where it pulled in: Cais de entrada

Whoever scans it counts the pallets. This only creates the task — and the clock starts now.

67 · A truck has arrived

Pôr a chegada de um camião dentro do sistema, que é o que cria a tarefa de picar.